25901271 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Core Course (DSC) - MG4DSCFTQ200 - FOOD CHEMISTRY

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer type

Answer any **5** questions

Each question carries **2** marks

1. Why do bread becomes rough and dry on storage? [A] / [CO2, CO3, CO5]
2. How do proteins act as emulsifying agents in food systems? [U] / [CO2, CO3, CO5]
3. Explain the concept of antioxidant synergism [U] / [CO1, CO2]
4. Differentiate between amylose and amylopectin. [An] / [CO2, CO3]
5. Differentiate between oil-in-water and water-in-oil emulsions with one example. [K] / [CO2, CO3, CO5]
6. Define caustic refining [K] / [CO4]
7. What is milk sugar? [U] / [CO2]

[2x5 = 10]

Part B

Short Essay Questions

Answer any **5** questions

Each question carries **4** marks

8.	What are non starch polysaccharides? Give it's application in food.	[U] / [CO2, CO3, CO5]
9.	Describe the effect of temperature, pH, metal ions and oxygen on enzymatic browning.	[U] / [CO2]
10.	Explain specific gravity of fats and oils and state it's significance in purity of fats and oils.	[U] / [CO2]
11.	Write in brief on the following physicochemical property of sugars (1) Crystallisation (2) Effect of heat	[U] / [CO2, CO3, CO5]
12.	Describe about the factors affecting activity of phenolase enzymes in foods.	[U] / [CO2]
13.	Define rancidity. Discuss about different types of rancidity.	[K] / [CO1, CO2]
14.	Discuss the application of pectin in gel formation.	[A] / [CO2, CO3, CO5]

[4x5 = 20]

Part C

Long Essay Questions

Answer any **2** questions

Each question carries **10** marks

15.	Describe the different methods and products of starch hydrolysis and its important applications in food.	[An] / [CO2, CO3, CO5]
16.	Define amylase enzymes and explain its types along with its functions.	[U] / [CO2]
17.	Explain the procedure to determine peroxide value with calculations and discuss it's importance in assessing rancidity.	[U] / [CO2]
18.	Discuss the structure and modifications of cellulose in food products.	[U] / [CO2, CO3, CO5]

[10x2 = 20]